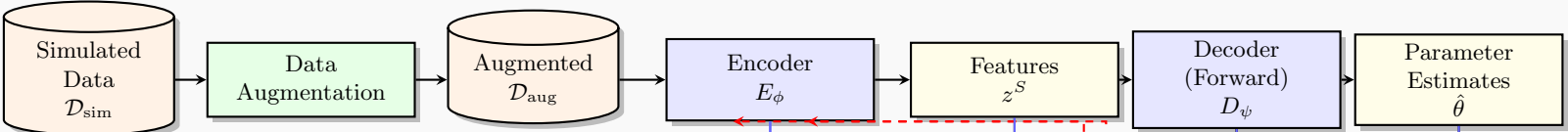
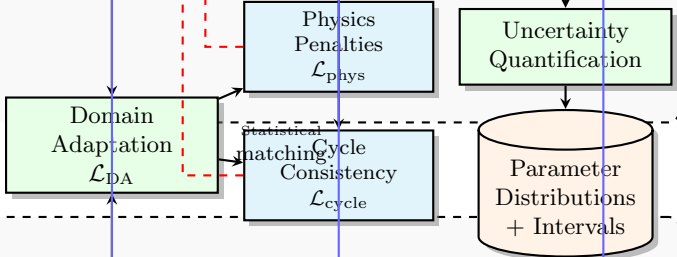
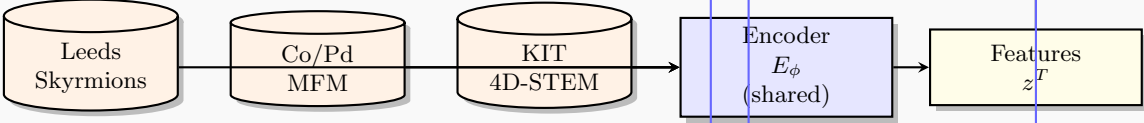


Simulated Domain



Experimental Domain



Training Loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}} + \gamma \mathcal{L}_{\text{phys}} + \delta \mathcal{L}_{\text{cycle}} + \lambda \mathcal{L}_{\text{DA}}$$

Interpretability & Validation



Key Components:

- Interpretability
- - - Feedback
- Data flow